

Customer Churn Analysis and Sales Insights Using Python

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250,000

PURCHASE RECORDS

12

DATA FIELDS

~20%

OVERALL CHURN RATE

0

DUPLICATE RECORDS

EXECUTIVE SUMMARY

This report summarizes the Customer Churn Analysis and Sales Insights project completed during my Data Analytics Internship at Alfido Tech. The project examines an e-commerce transaction dataset of 250,000 purchase records to understand customer churn, high-value customer behavior, and product category performance. Using Python, I cleaned and prepared the data, handled missing values in the Returns column, checked for duplicates, and engineered time-based fields before moving into exploratory data analysis. The analysis covers top customers by purchase volume, revenue split across product categories, quarterly sales trends, return patterns, churn behavior across gender and category, and correlation between key numerical variables. Overall churn stood at around 20%, with no duplicate records found in the dataset. The findings point to practical next steps for customer retention, inventory planning, and reducing product returns.

OBJECTIVE

- Analyze customer churn patterns
- Identify high-value customers
- Study product category performance
- Review return trends by category
- Turn findings into retention actions

DATASET OVERVIEW

- 250,000 rows · 12 columns
- Customer ID, Purchase Date, Category, Price, Quantity, Amount, Payment Method, Returns, Age, Gender, Churn
- No duplicate records
- ~19% missing Returns values, filled using mode

TOOLS & TECHNOLOGIES

- Python 3
- Pandas & NumPy for data handling
- Matplotlib & Seaborn for visualization
- Jupyter Notebook

WORKFLOW

Data loading and inspection (head, tail, info, describe, shape) → data cleaning (data type conversion, duplicate check, missing value treatment) → exploratory data analysis (segregating numerical/categorical columns, extracting month, year, and quarter from purchase date) → visualization of trends across customers, categories, quarters, returns, and churn → interpretation of results into business insights.

Code Repository: github.com/anshilxp1/Customer-Churn-Analysis-in-E-commerce-Using-Python

KEY FINDINGS

- A small group of customers accounts for a large share of purchase volume — a good target for loyalty programs.
- Revenue is not evenly spread across product categories; some categories consistently outperform others.
- Quantity sold shifts across quarters for different categories, pointing to seasonal demand that matters for inventory planning.
- Return rates vary by category — categories with higher returns are worth reviewing for quality or fit issues.
- Churn (~20% overall) is not evenly split across gender and category combinations, so targeted retention works better than a single blanket strategy.
- Price, quantity, amount, and age show only weak correlation, suggesting churn is driven more by category or service factors than by price alone.

VISUALIZATIONS

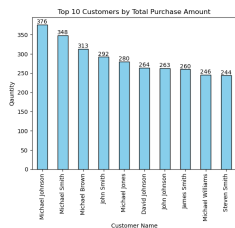


Fig 1. Top 10 Customers by Purchase Volume

A small set of customers drives a large share of quantity purchased.

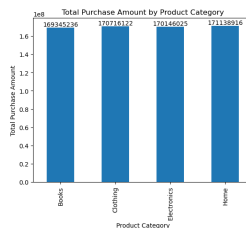


Fig 2. Revenue by Product Category

Purchase amount varies by category, with one category leading total revenue.

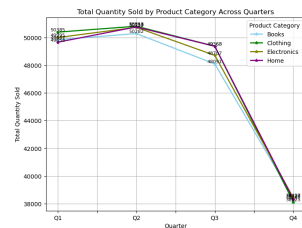


Fig 3. Quarterly Sales Trend by Category

Quantity sold shifts across quarters, indicating seasonal demand patterns.

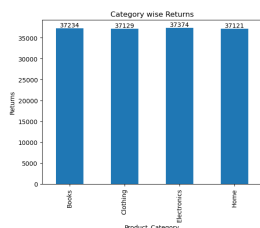


Fig 4. Category-wise Returns

Return volume differs by category, flagging where quality checks help most.

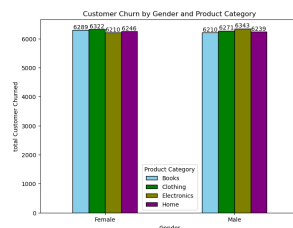


Fig 5. Churn by Gender and Category

Churn is uneven across gender-category pairs, supporting targeted retention.



Fig 6. Correlation Heatmap

Price, quantity, amount, and age show mostly weak correlation with each other.

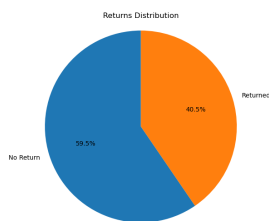


Fig 7. Returns Distribution

Roughly half of all purchases were returned, making returns a real cost driver.

WHAT THESE CHARTS TELL US

Revenue, returns, and churn are all concentrated rather than evenly spread — a handful of customers and categories carry most of the weight, and close to half of all orders come back as returns. That combination points to a segment-level approach for retention and quality checks, rather than broad, one-size-fits-all changes.

CONCLUSION

Working on this project helped me apply the full data analytics workflow, from cleaning a raw dataset to drawing insights that matter for business decisions. I got hands-on practice with Pandas for data manipulation and Matplotlib/Seaborn for visual storytelling, and learned how to turn charts into insights a business team can act on — who to retain, which categories need attention, and where returns are hurting margins. It also strengthened my confidence in handling real-world, imperfect data and presenting findings clearly for a non-technical audience.